

# Marko Mladenovic

marko@bugdigger.com | bugdigger.com | github.com/bbugdigger.com

## Work Experience

|                                                                                                                                                                                                                                                                                                         |                                      |                          |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------|--------------------------|
| <b>Game Integrity</b>                                                                                                                                                                                                                                                                                   | <b>Shindiri Studio – Nis, Serbia</b> | <b>10/2024 - Present</b> |
| <ul style="list-style-type: none"><li>Tested security of game clients, servers, mobile and web apps, cloud infrastructure and 3<sup>rd</sup> party integrations</li><li>Wrote polymorphic engine for hardening reverse engineering</li><li>Internally documented everything I did and learned</li></ul> |                                      |                          |
| <b>Cheat Developer</b>                                                                                                                                                                                                                                                                                  | <b>GameVandal</b>                    | <b>10/2023 – 10/2024</b> |
| <ul style="list-style-type: none"><li>Developed fully external cheats with kernel driver for CSGO and CS2</li><li>Worked on whole cheat deployment workflow</li><li>Optimized cheat code for maximum fps</li></ul>                                                                                      |                                      |                          |
| <b>Reverse Engineer Internship</b>                                                                                                                                                                                                                                                                      | <b>AST – Nis, Serbia</b>             | <b>05/2024 – 08/2024</b> |
| <ul style="list-style-type: none"><li>Reverse engineered and found vulnerabilities on a IP camera's firmware</li><li>Developed firmware for ARM64 in microC</li></ul>                                                                                                                                   |                                      |                          |
| <b>Backend Developer</b>                                                                                                                                                                                                                                                                                | <b>FI3xx GmbH – Vienna, Austria</b>  | <b>06/2022 – 07/2023</b> |
| <ul style="list-style-type: none"><li>Backend developer (Tech stack: Java, Spring, Hibernate, JPA, Elasticsearch, Postgres, Redis)</li><li>Maintained and expanded existing code base with occasional new feature development</li></ul>                                                                 |                                      |                          |

## Certifications & Achievements

- **Ret2 Systems Binary Exploitation Course** - [wargames.ret2.systems/course](https://wargames.ret2.systems/course) **February 2024**
  - Reverse engineered an algorithm that generates a software license
  - Found race condition vulnerability in concurrent garbage collector. Challenge was inspired by the JavaScriptCore vulnerability presented at Pwn2Own 2018 competition
- **ZDE VR Course** - [zerodayengineering.com/training/universal-vulnerability-research.html](https://zerodayengineering.com/training/universal-vulnerability-research.html) **February 2024**
  - Got essential knowledge and skills required to conduct modern low-level security research, vulnerability exploit development, secure software engineering and hardening

## Skills

- Programming languages (Java, C/C++, Rust, JavaScript, Python, x86-64 and ARM64 assembly)
- Reverse engineering, static analysis (ghidra, binary ninja, ida), and dynamic analysis (x64dbg, windbg)
- Ability to bypass common exploitation techniques (ASLR, DEP, PIE, Stack cookies)
- Exploit dev, writing exploits for userland and kernel (Windows, Linux)
- Systems programming in context of writing kernel drivers, compilers and hypervisors
- Malware analysis (unpacking, API resolving, emulation, VMProtect, Themida)
- Knowledge of cheat development process (internal and external) and detection vectors
- Experience with developing/reversing obfuscation, encryption and anti-tamper technologies
- Familiarity with game engine internals (Source2, Unreal, Unity) as well as game development design patterns

## Projects

- **BanHammer** - <https://github.com/bbugdigger/BanHammer> **In Progress**
  - Actively developing an open-source anti-cheat to explore common cheat detection vectors and replicate production level anti-cheat systems architecture
- **Rust ARM64 kernel for Raspberry Pi 3** - <https://github.com/bbugdigger/armRustOS> **In Progress**
  - Developing ARM64 kernel in rust for Raspberry Pi 3
- **Call of Duty Zombies Hack** - <https://github.com/bbugdigger/Call-of-Duty-WaW-Hack> **December 2023**
  - Reversed engineered a game and used dll injection to insert a hack into the game process
  - Hack uses function hooking on the D3D9 library graphics function